

Unifying Isolated Spin-Down and Binary Recycling in Massive Neutron Star Formation via Relaxed Equation of State Axioms

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Abstract—This research plan introduces a candidate mechanism for massive neutron star formation by formally relaxing the static equilibrium axioms of the dense matter equation of state to incorporate rotational and thermal support from binary mass accretion. While existing observational constraints—such as the radius and mass bounds derived from GW170817—demonstrate the necessity of a sufficiently stiff equation of state to support massive remnants, standard static Tolman-Oppenheimer-Volkoff (TOV) limits remain contested by the population of millisecond pulsars exceeding 2.0 solar masses. The proposed framework unifies isolated spin-down and binary recycling pathways by modeling mass transfer as a continuous thermodynamic and mechanical perturbation. This modification predicts a distinct mass-spin relation, formalized as $M = R(P)$, which differentiates binary-recycled massive neutron stars from isolated magnetic-dipole spin-down pulsars. The derivation is grounded in the assumption that companion ablation affects only the surface and magnetosphere, leaving the core governed by the relaxed effective equilibrium axioms.

The validity of the proposed mechanism is subject to strict boundary conditions, specifically the preservation of the causality bound ($c_s \leq c$) and consistency with the observed population of massive millisecond pulsars. The research design prespecifies outcome branches that distinguish between support for the relaxed axioms and the sufficiency of standard rotational support models. A null or contradictory outcome would indicate that the relaxed static equilibrium axioms inevitably violate causality or tidal deformability constraints, or that standard stiff equations of state fully account for the mass-spin relation without requiring axiom relaxation. Consequently, the study advances a testable formal bridge between microphysical nucleosynthesis networks and macroscopic remnant properties, awaiting proof obligation verification to determine if the relaxed axioms can consistently produce observed massive neutron star masses within the defined validity domain.

Index Terms—neutron stars, equation of state, binary recycling, mass-spin relation, causality, TOV limit

I. INTRODUCTION

A. Scientific Setting and Observational Tension

The existence of massive neutron stars, particularly millisecond pulsars exceeding 2.0 solar masses, places stringent constraints on the equation of state (EoS) of dense matter. Standard binary evolution models attribute these high masses to a combination of a stiff EoS and rotational support acquired through binary spin-up, allowing remnants to exceed the static Tolman-Oppenheimer-Volkoff (TOV) limit. However, a critical gap remains in the mechanistic explanation of how these objects maintain stability without violating causality bounds or tidal deformability constraints. While gravitational-wave observations and pulsar timing provide robust empirical anchors

for mass and spin parameters, the theoretical bridge linking binary accretion history to the effective equilibrium conditions of the neutron star core is underdeveloped. Current frameworks often treat rotational support as a post-hoc correction rather than a fundamental modification of the equilibrium axioms themselves. [1], [2]

B. Proposed Contribution: Relaxed Equilibrium Axioms

This research plan proposes a candidate mechanism that formally relaxes the static equilibrium axioms of the dense matter EoS to incorporate rotational and thermal support from binary mass accretion as continuous thermodynamic perturbations. The central hypothesis posits that this relaxation permits mass accumulation beyond the static TOV limit while preserving the causality bound ($c_s \leq c$). By modeling binary accretion as a modification to the effective equilibrium conditions, rather than merely an angular momentum transfer event, this approach aims to unify isolated spin-down and binary recycling pathways. The proposed contribution is a distinct mass-spin relation, $M = R(P)$, which predicts that massive neutron stars ($> 2.0 M_{\text{sun}}$) with millisecond periods follow a specific evolutionary track distinguishable from isolated magnetic-dipole spin-down pulsars.

C. Theoretical Design and Verification Strategy

The study employs a mathematical/theoretical design, treating the theoretical model as the experimental unit. The verification strategy relies on formal proof obligations to demonstrate that the relaxed axioms do not inevitably violate causality or tidal deformability constraints. Counterexample analysis and numerical verification plans are designated to test the predicted mass-spin relation against the observed population of massive millisecond pulsars. The design assumes that companion ablation primarily affects the surface and magnetosphere, allowing the core mass-spin relation to remain governed by the relaxed effective equilibrium axioms. This separation of formal reasoning from empirical comparison ensures that the proposal remains grounded in theoretical consistency while awaiting observational validation.

II. BACKGROUND, SURVEY, AND RESEARCH GAP

A. Massive Neutron Stars and the Limits of Static Equations of State

The existence of massive neutron stars provides a stringent boundary condition for any theory of dense matter and stellar

evolution. Radio-timing measurements of binary millisecond pulsars, specifically systems like PSR J1614-2230, demonstrate gravitational masses approaching two solar masses. These observations require an equation of state (EoS) for dense matter that is sufficiently stiff at supranuclear densities to prevent gravitational collapse. However, standard static Tolman-Oppenheimer-Volkoff (TOV) models struggle to reconcile these high masses with the simultaneous requirement of causal sound speeds and constraints from gravitational-wave events. The challenge is not merely numerical but structural: static equilibrium models assume spherical symmetry and non-rotation, whereas the observed massive remnants are rapidly spinning millisecond pulsars. This discrepancy necessitates a theoretical framework that explicitly incorporates rotational and thermal support into the equilibrium conditions, moving beyond the limitations of static TOV solutions. [1], [3]

B. Rotational Support and Binary Recycling Dynamics

Binary recycling offers a mechanism to explain the high spin frequencies of these massive remnants. Accretion from a companion star transfers angular momentum, spinning up the neutron star and potentially increasing its maximum stable mass through centrifugal support. Population synthesis models indicate that binary interactions fundamentally alter progenitor trajectories, modulating mass loss and rotation rates. However, current modeling frameworks often treat the EoS and the rotational state as decoupled parameters. While the stiffening effect of rotation is acknowledged, it is typically applied as a post-hoc correction to static mass limits rather than being integrated into the fundamental equilibrium axioms. This separation obscures the precise dynamical conditions under which a neutron star can sustain masses beyond the static TOV limit without violating causality or triggering instability. The literature confirms that binary mass transfer efficiency and magnetic braking are major sources of uncertainty, further complicating the link between accretion history and the final mass-spin state. [4], [5]

C. The Mechanistic Gap in Relaxed Equilibrium

A critical mechanism_explanation_gap persists in the relevant evidence cluster concerning massive neutron star masses. The survey evidence confirms that the 'candidate_mechanism' slot remains uncovered. While we observe massive millisecond pulsars and we possess static EoS models, there is no established formal theory that relaxes the static equilibrium axioms to incorporate rotational and thermal support in a way that preserves causality. Existing alternatives rely on standard stiff EoS combined with rotational support, but these models do not formally modify the underlying equilibrium conditions of the dense matter itself. Consequently, the field lacks a predictive mass-spin relation that distinguishes the evolutionary channel of binary recycling from isolated spin-down. This gap prevents a causal interpretation of how massive remnants form and survive, leaving the connection between microphysical EoS properties and macroscopic binary evolution underconstrained.

D. Missing Premises for Causal Consistency

The proposed contribution addresses this gap by introducing a formal relaxation of static equilibrium axioms. This theoretical move requires specific missing premises to be established. First, the mathematical form of the relaxation must be defined such that it permits mass accumulation beyond the static limit without violating the speed-of-light constraint for sound propagation. Second, the continuous thermodynamic and mechanical perturbation caused by binary accretion must be modeled as a modification to the effective equilibrium conditions, not just an external torque. Third, the interaction between companion ablation and the core mass-spin relation must be characterized to ensure that surface effects do not invalidate the core equilibrium assumptions. These premises are currently absent from the frozen evidence, representing the precise intellectual labor required to bridge the gap between observed massive pulsars and theoretical dense matter physics.

E. Transition to Formal Argument

The next stage of the research plan will operationalize these missing premises into a formal argument. We will define the candidate theorem entry, specifying the domain of admissible premises for the relaxed equilibrium axioms. This will involve establishing a dependency-closure matrix that maps the rotational support mechanisms to the causality bounds. By formalizing these relationships, we aim to derive a distinct mass-spin relation that serves as a discriminative criterion for binary recycling channels. The subsequent sections will detail the proof obligations and falsification criteria necessary to validate this theoretical framework against the observed population of massive neutron stars.

III. RESEARCH QUESTIONS AND PLANNED CONTRIBUTIONS

A. Research Questions

The proposed mechanism replacement targets two operational questions. First, what exact mathematical relaxation conditions permit static equilibrium axioms of dense matter equations of state to support masses beyond the static Tolman-Oppenheimer-Volkoff limit without violating the causality bound? Second, what numerical or analytical techniques verify the predicted mass-spin relation that distinguishes massive millisecond pulsars from isolated magnetic-dipole spin-down systems? These questions are delimited by unresolved design dependencies requiring human confirmation. [3], [4]

B. Planned Contributions

This research plan contributes a candidate mechanism unifying isolated spin-down and binary recycling via relaxed equation of state axioms. It formalizes the relation between gravitational mass and spin period for massive neutron stars exceeding the static limit. The contribution relies on binary mass accretion and angular momentum transfer modeled as a continuous thermodynamic and mechanical perturbation, distinguishing evolutionary channels through derived model outputs. [3]

C. Design Assumptions and Scope

The design assumes that companion ablation primarily affects the surface and magnetosphere, allowing the core mass-spin relation to remain governed by the relaxed effective equilibrium axioms. The scope is strictly bounded by the causality constraint and the requirement that standard rotational support models do not fully explain the observed population of massive millisecond pulsars without relaxing the static axioms. [4]

D. Transition to Formalization

The formalization of these questions and contributions is deferred to the definitions and propositions ledger, which owns the complete definition of the relaxed axioms and the candidate theorem entry. [3]

IV. IDEA SOURCE CHECKPOINTS AND DIRECTION SELECTION AUDIT

A. Origin and Scientific Attraction of the Selected Direction

The selected direction, titled "Unifying Isolated Spin-Down and Binary Recycling in Massive Neutron Star Formation via Relaxed Equation of State Axioms," emerges from the survey's identification of a critical mechanistic gap: the absence of a causal pathway linking dense matter equation of state (EoS) constraints to the observed population of massive millisecond pulsars. The scientific attraction of this proposal lies in its ambition to resolve the tension between static Tolman-Oppenheimer-Volkoff (TOV) limits and the existence of neutron stars exceeding 2.0 solar masses. By positing that rotational and thermal support from binary accretion can formally relax static equilibrium axioms, the direction offers a unified theoretical framework that distinguishes binary-recycled remnants from isolated magnetic-dipole spin-down pulsars. This approach directly addresses the survey's finding that current models rely on phenomenological descriptions rather than predictive, causally grounded mechanisms for massive remnant formation. The proposal asserts that a distinct mass-spin relation, derived from these relaxed axioms, provides the necessary discriminator to separate evolutionary channels, thereby transforming a descriptive cataloging problem into a testable mathematical theory. [1], [2]

B. Defects and Unsupported Links Exposed by Checkpoints

- Audit snapshots from the idea evolution phase reveal significant formal defects that constrain the current viability of the selected direction. First, the specific mathematical relaxation conditions required to modify the static equilibrium axioms are referenced but not formally defined, leaving the core mechanism abstract. Second, the functional form of the proposed "mechanism_mass_spin_relation" remains unspecified, preventing the derivation of concrete predictions for mass-spin correlations. Third, the operational definitions for key variables, including the measurement method for masses greater than 2.0 solar masses and the precise criteria for testing causality violations, are missing. These gaps expose an unsupported link between the theoretical premise

of relaxed axioms and the observable prediction of a distinct mass-spin relation. Without formal definitions for the perturbation terms introduced by binary accretion, the proposal cannot yet demonstrate that the relaxed axioms inevitably avoid causality violations or tidal deformability constraints, a failure condition explicitly noted in the boundary analysis.

C. Qualified Retained Direction and Explicit Exclusions

Despite these formal defects, the direction is retained as a candidate mechanism because it correctly identifies the need for a unified theoretical treatment of rotational support in massive remnants. The retained scope is qualified by the exclusion of any claim that the relaxed axioms have been proven or that the mass-spin relation has been observed. The proposal explicitly excludes the possibility that standard stiff EoS combined with conventional rotational support fully explains the observations without relaxing fundamental nuclear EoS axioms; this alternative explanation remains a competing hypothesis that the proposed mechanism must eventually distinguish itself from. The current design assumes that companion ablation primarily affects the surface and magnetosphere, allowing the core mass-spin relation to remain governed by the relaxed effective equilibrium axioms. This assumption is a design premise, not a verified outcome. The transition to the next stage requires the formal definition of the relaxation conditions and the derivation of the mass-spin relation to test whether the predicted distinct track can be separated from isolated pulsar evolution.

V. PROBLEM DEFINITION, ASSUMPTIONS, AND HYPOTHESES

A. Formal Problem and Admissible Domain

The formal problem is to determine whether a candidate mass-spin relation for massive neutron stars can be derived from relaxed static equilibrium axioms while preserving causal constraints. The admissible domain comprises a theoretical model representing a neutron star in static equilibrium, governed by a dense matter equation of state and subject to external perturbations from binary accretion. This domain is delimited by the requirement that the speed of sound in the dense matter remains subluminal. The problem is not to validate observational data, but to establish a formal logical structure that maps relaxed axioms to a distinct evolutionary track. The survey identifies a critical gap in mechanistic explanation for massive neutron star formation, specifically within the context of dense matter equations of state. This section supplies the scoped premises for the subsequent derivation of the mass-spin relation.

B. Core Formal Definitions

a) *Definition.*: The domain is defined by the following variables and symbols: Let M represent the gravitational mass of the neutron star. Let P represent the neutron star spin period. Let R denote the derived mechanism_mass_spin_relation function such that $M = R(P)$. Let c_s represent the local speed

of sound in the dense matter, and c represent the speed of light. The causal speed limit check is defined as the constraint $c_s \leq c$. The evolutionary channel is defined as a binary variable distinguishing isolated magnetic-dipole spin-down from binary recycling. The static equilibrium axioms are defined as the set of conditions governing hydrostatic balance in the absence of rotational support.

C. Assumption Ledger

- The formal problem relies on the following candidate assumptions, which require human review to confirm their mathematical consistency: 1. The static equilibrium axioms of the dense matter equation of state can be formally relaxed to permit mass accumulation beyond the static Tolman-Oppenheimer-Volkoff limit via rotational and thermal support without violating the causality bound. 2. Binary mass accretion and angular momentum transfer can be modeled as a continuous thermodynamic and mechanical perturbation that modifies the effective equilibrium conditions of the neutron star. 3. Companion ablation primarily affects the surface and magnetosphere, allowing the core mass-spin relation to remain governed by the relaxed effective equilibrium axioms.

D. Numbered Problem Relation

$$M = R(P) \quad (1)$$

E. Relation Interpretation

The relation $M = R(P)$ states that the gravitational mass is a function of the spin period under the relaxed axioms. This relation is the central object of the formal problem. The specific functional form of $R(P)$ is not defined in the upstream artifacts and requires formal specification. The relation serves as the bridge between the mechanical perturbation of binary accretion and the observable properties of the neutron star. See Eq. (1).

F. Candidate Propositions

a) Proposition (Candidate): Proposition P1: Relaxing static equilibrium axioms of dense matter EoS with rotational/thermal support permits masses beyond static TOV without violating causality. Proposition P2: The relaxed axioms predict a distinct mass-spin relation distinguishing massive NSs ($\gtrsim 2.0$ Msun) with ms periods from isolated magnetic-dipole spin-down pulsars.

G. Candidate Lemma L1

a) Lemma L1 (Candidate): Lemma L1: Under Assumptions A1, A2, and A3, the static equilibrium constraint is modified by a continuous perturbation term associated with binary accretion support. This lemma is a candidate formalization derived from Proposition P1 and requires verification of proof obligation PO1.

H. Candidate Lemma L2

a) Lemma L2 (Candidate): Lemma L2: Under Assumptions A1, A2, and A3, a derived mass-spin relation $R(P)$ exists for the relaxed axioms. This lemma is a candidate formalization derived from Proposition P2 and requires verification of proof obligation PO2.

I. Proof Obligations and Failure Conditions

The candidate propositions are subject to proof obligations PO1 and PO2. PO1 requires demonstrating that the relaxation of static equilibrium axioms preserves the causality bound. PO2 requires demonstrating that the derived relation $R(P)$ yields masses greater than the static maximum for millisecond periods. The claim fails if the relaxed static equilibrium axioms violate the causality bound, or if the predicted mass-spin relation contradicts the observed population of massive millisecond pulsars, or if standard rotational support models fully explain the observations without needing the relaxed axioms.

VI. STUDY DESIGN AND METHODS

The unit of analysis is a static theoretical model in which the equilibrium axioms of the dense-matter equation of state are relaxed to include rotational and thermal support from binary accretion. The protocol specifies three independent axes: the binary accretion parameters, the causality bound, and the equilibrium conditions. The dependent outputs are gravitational mass and spin period, reported in solar masses and milliseconds. The frozen formal reasoning plan supplies the candidate propositions and a six-step forward derivation; this section consumes that structure to define the computational protocol without re-listing its unresolved symbols. The derivation begins by applying the relaxation to the static equilibrium constraint, then derives a candidate mass-spin relation, evaluates the mass limit at millisecond periods, checks the causality bound, and finally contrasts the resulting track with isolated magnetic-dipole spin-down. The protocol inherits the status of each step from the formal plan: the first step is proposed and the remaining steps are unverified, so the design is a verification schedule, not an established result. [3]

The design tests the relaxed-axiom mechanism against a declared alternative in which standard stiff equations of state plus rotational support from binary spin-up already account for massive millisecond pulsars without any relaxation of the fundamental nuclear axioms. The comparison is therefore a model-competition protocol: the relaxed-axiom track must produce a mass-spin relation that the standard-support track cannot reproduce within the same causal bound. Ablations isolate each control axis. Removing the rotational and thermal support term collapses the model to the static limit, and tightening the causality bound tests whether the predicted masses survive the speed-of-sound constraint. Robustness is treated as a numerical-implementation problem. The MESA binary module supplies the computational precedent for differentially rotating stars and implicit mass-transfer schemes, but it explicitly excludes common-envelope and contact-binary

phases, so those phases are declared out of scope rather than silently absorbed. The Aquila comparison project demonstrates that code-to-code variation in subgrid physics can dominate the result even at fixed initial conditions; the design consequently requires a cross-implementation check before any mass-spin relation is treated as model-independent. [5], [6]

[1], [3]

a) Planned protocol.: The verification protocol maps each prespecified outcome branch to an allowed conclusion. A result consistent with the declared relation while the planned controls do not favor the standard-support alternative yields the `supports_mechanism` branch, which supports but does not prove the relation within the design boundary. Variation across declared conditions, units, or measurement contexts yields `partial_or_heterogeneous` and forbids a universal conclusion. A comparison that does not support the relation, or that favors the declared alternative, yields `null_or_contradictory`; absence of support is not proof of absence generally. A failure of quality control, protocol deviation, or validity criterion yields `uninformative_or_invalid` and warrants no scientific conclusion until the prerequisite is resolved. The counterexample analyzer batch was discarded and is not retained, so no counterexample is claimed; the boundary analysis above is the design-level substitute until qualified human review replaces it. The protocol consumes the formal plan’s proof obligations as verification targets rather than treating them as completed proofs.

b) Human-review checklist.: The MESA infrastructure provides bit-for-bit consistency across supported platforms, and the design adopts that standard as the reproducibility target for the numerical implementation. The measurement plan, quality-control procedure, data-management policy, and reproducibility protocol are not yet finalized and require human input before the design is treated as executable; the measurement plan in particular must specify how gravitational mass and spin period are operationalized, including the method for masses above 2.0 solar masses and the timing procedure for millisecond periods. These four unresolved records are the only human-review dependencies owned by this section, and they gate the transition from the design protocol to the verification stage. The detailed definition ledger and the risk-and-review ledger are owned elsewhere and are referenced here only as the sources of the unresolved symbols and the review gates. [5]

VII. EXPECTED OUTCOME BRANCHES AND CONDITIONAL CONCLUSIONS

A. Scope of the Decision Protocol

This route owns the conditional-outcome protocol for a candidate mechanism that has not yet produced observed results. The two propositions under test are: relaxing static equilibrium axioms of the dense matter equation of state with rotational and thermal support permits masses beyond the static Tolman-Oppenheimer-Volkoff limit without violating causality, and the relaxed axioms predict a distinct mass-spin relation distinguishing massive neutron stars exceeding 2.0

solar masses with millisecond periods from isolated magnetic-dipole spin-down pulsars. The decision matrix below maps each prespecified branch to the relevant proof obligation, the conclusion the branch licenses, and the next action. Established upstream evidence constrains the design but does not confirm the mechanism: gravitational-wave analyses show that neutron-star radius and component-mass constraints depend on whether the equation of state is required to support masses above 1.97 solar masses and on spin-magnitude priors, while heavy-ion and astrophysical joint analyses indicate that dense-matter pressure constraints tighten when microscopic collision data are combined with macroscopic observations. These facts set the comparison baseline against which the relaxed-axiom relation must differentiate itself. [1], [7], [8]

B. How Each Branch Updates the Argument

a) Pre-registered Branch (Expected—Not Observed).:

The supportive branch is the only one that advances the candidate mechanism, and it does so strictly as a within-boundary support: a consistent mass-spin relation that survives the standard stiff-equation-of-state rotational-support control would favor the relaxed axioms without establishing them. The null or contradictory branch is decisive in the opposite direction, because the declared failure condition is met when the relaxed axioms violate the causality bound, when the predicted relation contradicts the observed massive millisecond-pulsar population, or when standard rotational support fully accounts for the observations; in that case the competing explanation prevails and the mechanism claim is not supported here. The uninformative or invalid branch is a procedural outcome, not a scientific one: it withholds any theorem-status update and routes the design back to human confirmation of the measurement, sampling, and analysis prerequisites rather than recording a failure of the proposal.

C. Transition to Review

The matrix consumes the scoped premises carried forward from the derivation stage and hands the resulting conditional conclusions to the review ledger, where the unresolved formal symbols, the discarded counterexample batch, and the open operational definitions are adjudicated. No branch in this protocol is treated as observed; each is an expected outcome that either licenses a bounded conclusion or defers to human review.

VIII. RISKS, LIMITATIONS, AND HUMAN REVIEW REQUIREMENTS

A. Scope of Limitations and Review Obligations

The proposed mechanism for relaxing static equilibrium axioms in dense matter equations of state to accommodate rotational and thermal support remains a candidate theoretical framework. No empirical results have been observed, and the formal reasoning chain is explicitly marked as requiring qualified human review. The primary limitation is the absence of a defined mathematical relaxation condition and an unverified derivation of the mass-spin relation. These gaps prevent the

TABLE I
DECISION MATRIX

Condition	Design role	Consequence if violated
Causality bound ($c_s \neq c$)	Hard control on every candidate equation of state	The relaxed axioms are rejected as physically inadmissible
Static TOV limit	Reference baseline the relaxation must exceed	No mass-accumulation claim beyond the static maximum
Massive MSP population ($M \geq 2.0 M_{\text{sun}}$, ms periods)	Empirical consistency target	The predicted mass-spin relation contradicts the observed population
Standard rotational support	Declared alternative explanation	The relaxation is unnecessary and the mechanism is not favored
Common-envelope / contact phases	Declared out of scope	Results cannot be extended into these regimes

TABLE II
CONDITIONAL-OUTCOME DECISION MATRIX

Branch	Prespecified trigger	Relevant obligation	Allowed conclusion	Next action
supports_mechanism	Prespecified analysis is consistent with the declared relation while planned controls do not favor a stated alternative explanation.	PO1, PO2	Supports, but does not prove, the relaxed-axiom mass-spin relation within the design boundary.	Replicate under independently confirmed conditions; test the most consequential declared boundary condition.
partial_or_heterogeneous	Analysis indicates variation across declared conditions, units, or measurement contexts.	PO1, PO2	The relation is conditional or heterogeneous; no universal conclusion is warranted.	Predefine and check plausible moderators; improve coverage of conditions and measurement comparability.
null_or_contradictory	Comparison does not support the declared relation or favors the standard rotational-support alternative.	PO1, PO2	The proposed relation is not supported in this design boundary; absence of support is not proof of general absence.	Audit construct validity and comparison adequacy; revise the mechanism or boundary claim before another design iteration.
uninformative_or_invalid	Quality-control, missingness, protocol-deviation, or validity criteria prevent interpretation.	PO1, PO2	No scientific conclusion is warranted; theorem status does not update.	Resolve the identified validity or data-quality failure; obtain human confirmation of measurement, sampling, and analysis prerequisites.

elevation of the central hypothesis to a validated theoretical result. The review requirements focus on formalizing the equilibrium perturbation terms, verifying causality bounds, and replacing the discarded counterexample analysis batch with qualified expert input.

B. Handling of No-Information Branches

a) Decision Status: No-information.: The theory spine identifies multiple no-information branches associated with undefined definitions and unverified proof obligations. These branches indicate that the dependency does not update theorem status; they are neither proof of failure nor a reason to describe the whole research plan as invalid. The response to each no-information branch is to resolve or review the upstream input. Specifically, the lack of formal definitions for the relaxed axioms and the causality check prevents the proof obligations from being discharged, requiring human intervention before any theoretical claim can be considered established.

C. Concrete Human-Review Decisions

a) Human-review checklist.: 1. Define the exact mathematical relaxation conditions required for the static equilibrium axioms to permit mass accumulation beyond the static Tolman-Oppenheimer-Volkoff limit. 2. Specify the numerical methods or analytical techniques to verify the derived

mass-spin relation. 3. Replace the discarded counterexample_analyzer batch with qualified human review before treating the design as complete. 4. Confirm that the relaxed static equilibrium axioms do not violate the causality bound under all admissible premises.

IX. DEFINITIONS, PROPOSITIONS, AND PROOF OBLIGATIONS

A. Symbol Ledger and Domain Conventions

a) Decision Status: No-information.: This section owns the formal symbol domain for the candidate mechanism. The gravitational mass M and spin period P are operationalized in solar masses and milliseconds respectively, serving as the dependent variables for the mass-spin relation. The causal speed limit check c enforces the dimensionless constraint $c_s \neq c$ on the dense matter equation of state. The evolutionary channel A3 distinguishes between isolated magnetic-dipole spin-down and binary recycling. The relaxed static equilibrium axioms A1 and the binary accretion support mechanism A2 act as the primary independent perturbations. The derived relation $R(P)$ maps the spin period to the gravitational mass. These symbols form the basis for the candidate propositions and proof obligations, with their operational definitions requiring

TABLE III
LIMITATION AND REVIEW DECISION MATRIX

Domain	Limitation / Missing Premise	Review Requirement	Release Condition
Formal Definitions	The specific mathematical relaxation of static equilibrium axioms (A1) is undefined.	Qualified human review of formal reasoning plan.	Explicit definition of perturbation terms for rotational/thermal support.
Counterexamples	Counterexample analysis was not run; no conclusion can be drawn.	Replace discarded counterexample_analyzer batch with human review.	Validated counterexample or boundary analysis for the relaxed axioms.
Causality Bounds	Formal criteria for testing causality violation ($c_s \neq c$) are missing.	Human review of causality check implementation.	Proof or numerical verification that relaxation preserves causality.
Mass-Spin Relation	The functional form of mechanism_mass_spin_relation (R) is undefined.	Human review of derivation steps S2-S6.	Derived relation consistent with observed massive millisecond pulsars.
Workflow Degradation	LLM or workflow degradation triggered review requirements.	Review of risk gate and degradation triggers.	Confirmation that all upstream formal and empirical claims remain separate.

human review to resolve the specific mathematical relaxation conditions.

$$M = R(P, A1, A2, A3) \quad (2)$$

The numbered relation defines the candidate mass-spin function R. It posits that the gravitational mass M is a function of the spin period P, modulated by the relaxed equilibrium axioms A1, the binary accretion support A2, and the evolutionary channel A3. This relation is the central object of the formal derivation. The specific functional form of R is currently undefined and requires formal specification, as noted in the unresolved inputs. See Eq. (2).

B. Candidate Lemma Registry

a) *Lemma Registry L1, L2, L3, L4, L5, L6, L7, L8 (Candidate).*: The theory spine defines eight candidate lemmas (L1-L8) that structure the forward derivation. L1 and L2 are candidate formalizations of the primary propositions P1 and P2. L3 represents the proposed application of the relaxation to the static equilibrium constraint. L4 through L8 are unverified lemmas that attempt to derive the existence of R(P), evaluate the mass limit, check the causality bound, assess the evolutionary channel, and distinguish the pulsar classes. These lemmas are strictly unverified and depend on the resolution of the formal definitions.

C. Candidate Propositions

a) *Proposition (Unverified).*: P1 proposes that relaxing the static equilibrium axioms of the dense matter equation of state with rotational and thermal support permits masses beyond the static Tolman-Oppenheimer-Volkoff limit without violating the causality bound. P2 proposes that these relaxed axioms predict a distinct mass-spin relation R(P) that distinguishes massive neutron stars ($M \gtrsim 2.0$ solar masses) with millisecond periods from isolated magnetic-dipole spin-down pulsars. Both propositions are candidate formalizations and remain unverified pending the resolution of the proof obligations.

D. Assumption Ledger

- The formal argument rests on three candidate assumptions:
- A1: Static equilibrium axioms can be relaxed to permit mass accumulation beyond the TOV limit via rotational and thermal support without violating causality.
- A2: Binary mass accretion and angular momentum transfer can be modeled as a continuous thermodynamic and mechanical perturbation modifying effective equilibrium conditions.
- A3: Companion ablation primarily affects the surface and magnetosphere, allowing the core mass-spin relation to remain governed by the relaxed effective equilibrium axioms.
- These assumptions require human review to establish their mathematical rigor.

X. FORWARD DERIVATION AND COUNTEREXAMPLE SEARCH PLAN

A. Formal Setup and Admissible Premises

a) *Definition.*: The derivation operates on the candidate premises inherited from the definition ledger: A1 (formal relaxation of static equilibrium axioms via rotational and thermal support), A2 (binary mass accretion and angular momentum transfer as a continuous thermodynamic and mechanical perturbation), and A3 (companion ablation confined to the surface and magnetosphere, preserving core mass-spin governance). The target object is the proposed function R(P), mapping the spin period P to the gravitational mass M. This section advances the argument by specifying the admissible comparators and the failure conditions under which the derivation is invalidated, without re-listing the full ledger entries.

B. Proposed mathematical dependency

$$M(P) = R(P; A1, A2, A3) \quad \text{\textit{subject to}} \\ \frac{dP}{dt} = f_{\text{\textit{spin-down}}}(P, B, \alpha, \dot{\alpha})$$

TABLE IV
DEPENDENCY AND FAILURE CONDITION MATRIX

Proof Obligation	Required Lemmas	Failure Condition	Status
PO1	L1, L3, L4, L6	Relaxed axioms violate causality bound ($c_s \not\leq c$)	needs_human_input
PO2	L2, L5, L7, L8	Predicted R(P) contradicts observed massive MSP population	needs_human_input

C. Derivation Relation and Perturbation

The numbered relation establishes the core derivation obligation: the relaxed equilibrium axioms must yield a mass function $R(P)$ that exceeds the static Tolman-Oppenheimer-Volkoff limit for millisecond periods, while the spin evolution is governed by the torque function. The static equilibrium constraint is modified by a continuous perturbation term associated with binary accretion support (L3). This relation is proposed, not observed, and its validity hinges on the explicit mathematical specification of the relaxation terms and the accretion perturbation.

D. Lemma Registry and Proof Obligations

a) Lemma Registry (Unverified).: Candidate L3 posits that the static equilibrium constraint is modified by the continuous perturbation term from binary accretion support. Unverified L4 asserts that a derived mass-spin relation $R(P)$ exists for the relaxed axioms. Unverified L5 requires that $R(P)$ yields $M \geq M_{\text{static_max}}$ for millisecond periods P . Unverified L6 mandates that the relaxation preserves the speed of sound constraint. Unverified L7 and L8 require that isolated magnetic-dipole spin-down pulsars evolve along a distinct track, separating them from the binary-recycled population. These lemmas are bound to proof obligations PO1 and PO2, which remain unverified and require formal completion.

E. Proposed Derivation Obligation

a) Proposition (Candidate).: Proposition P2 requires a formal proof that the relaxed static equilibrium axioms and the derived mechanism_mass_spin_relation consistently produce the observed massive neutron star masses within the massive_neutron_star_validity_domain. The specific mathematical relaxation conditions required for the static equilibrium axioms (A1) and the equations governing binary mass and angular momentum transfer (A2) are currently undefined. Consequently, the derivation obligation cannot be discharged until the exact mathematical relaxation conditions and the functional form of $R(P)$ are formally specified and verified.

F. Failure Conditions and Upstream Dependencies

- The derivation fails under the following prespecified conditions:
- The relaxed static equilibrium axioms violate the causality bound.
- The predicted mass-spin relation contradicts the observed population of massive millisecond pulsars.
- Standard rotational support models fully explain the observations without needing the relaxed axioms.

- These failure conditions are contingent on the resolution of upstream unknowns. The exact mathematical relaxation conditions for A1 and the equations for A2 require formal definition. The specific functional form of $R(P)$ is referenced but not defined. Until these premises are formally specified, the derivation remains an unverified proposal.

XI. COMPUTATIONAL EVIDENCE (NUMERICAL SIMULATION; NON-EMPIRICAL)

Q1 (final v0) answers: Can the joint distribution of braking index n and characteristic age τ_c statistically distinguish isolated magnetic-dipole evolution from binary recycling populations, given specific torque model constraints? Model family: Explicit Runge-Kutta 4th Order. Execution mode: NUMERICAL_SIMULATION. Result kind: SIMULATED. Empirical claim status: NOT_EMPIRICAL. Model-internal hypothesis relation: CONSTRAINED. Model-internal result summary: Within the audited reduced ODE, the isolated track increases from $P=0.500$ s to 0.662 s, whereas the recycling track reaches $P_{\text{min}}=0.226$ s during accretion and ends at 0.281 s after resumed dipole spin-down. This establishes model-internal directional separation, but does not by itself validate population-level n - τ_c discrimination because the executable MathIR contains two deterministic tracks rather than the documented Monte Carlo outer loop. Applicability conditions: Stellar geometry, electromagnetic constants, and moment of inertia are absorbed into effective braking constants K_{dipole} and K_{acc} ; Magnetic inclination evolves smoothly via exponential decay without discontinuous glitches or magnetospheric state transitions.; Accretion torque is active exclusively during $t \in [t_{\text{acc}}, t_{\text{dot}}]$ with nonzero $\dot{M}$; isolated dipole torque applies otherwise.. Limitations: Excludes glitch events, magnetospheric state transitions, and variable moment of inertia.; Assumes fixed stellar geometry and absorbs unresolved electromagnetic factors into effective constants.; Observational selection effects and survey detection thresholds are not dynamically coupled to the simulation loop.. Iteration lineage: v0: CONSTRAINED (Within the audited reduced ODE, the isolated track increases from $P=0.500$ s to 0.662 s, whereas the recycling track reaches $P_{\text{min}}=0.226$ s during accretion and ends at 0.281 s after resumed dipole spin-down. This establishes model-internal directional separation, but does not by itself validate population-level n - τ_c discrimination because the executable MathIR contains two deterministic tracks rather than the documented Monte Carlo outer loop.). Supplementary mathematical-model PDF: quantitative_mathematical_models.pdf#Q1.

TABLE V
COUNTEREXAMPLE AND BOUNDARY DECISION MATRIX

Condition	Classification	Required Response
Predicted $R(P)$ yields $M \leq 2.0 M_{\text{sun}}$ for $P \leq 10$ ms	Would-falsify	Reject the relaxed axioms; revert to standard stiff EoS with rotational support
Relaxation term violates $c_s \neq c$	Would-falsify	Invalidate A1; enforce strict causality bound without relaxation
Specific mathematical relaxation of A1 is undefined	Scope delimiter	Withhold theorem status update; resolve upstream formal definition
Functional form of $R(P)$ is unspecified	Scope delimiter	Withhold theorem status update; resolve upstream formal definition
Numerical implementation details are unspecified	No-information	Withhold theorem status update; obtain human confirmation of prerequisites
Standard rotational support fully explains observations	Would-falsify	Reject the necessity of A1; adopt alternative explanation

Q2 (final v0) answers: How does the surface temperature evolution differ between massive neutron stars undergoing spin-down induced core compression and those without, and can this distinguish the phase-transition mechanism from standard cooling? Model family: Stiff ODE IVP integrator (e.g., LSODA or Radau). Execution mode: NUMERICAL_SIMULATION. Result kind: SIMULATED. Empirical claim status: NOT_EMPIRICAL. Model-internal hypothesis relation: SUPPORTED_WITHIN_MODEL. Model-internal result summary: The audited reduced ODE crosses $\rho_{\text{crit}}=5.12e17 \text{ kg/m}^3$ at approximately $9.85e11 \text{ s}$ in all spin-down cases. At $3\tau_{\text{cool}}$, the standard, phase-transition, and phase-transition-plus-superfluid scenarios yield surface temperatures of about $4.98e4 \text{ K}$, $5.32e4 \text{ K}$, and $2.36e5 \text{ K}$, respectively; phase_fraction reaches 1 only in the two transition scenarios. The mechanism therefore produces distinguishable model-internal thermal tracks, conditional on the effective cooling, latent-heat, and compression proxies. Applicability conditions: Neutron stars are treated as spherically symmetric, isolated systems with no binary accretion torque or magnetic field evolution.; Detailed neutrino emissivity processes and heat-capacity microphysics are absorbed into a single effective cooling timescale and scenario-dependent coefficients.; Moment of inertia variations are fully parameterized into the rotational compression coefficient rather than solved dynamically.. Limitations: Highly sensitive to the choice of EOS and phase transition parameters; observational uncertainties in distance and radius affect temperature inference.; Assumes spherical symmetry and ignores binary recycling, companion-ablation evolution, and detailed crustal heat blanketing.; No resolved neutrino microphysics or full equation-of-state integration; relies on effective proxy coefficients for first executable run.. Iteration lineage: v0: SUPPORTED_WITHIN_MODEL (The audited reduced ODE crosses $\rho_{\text{crit}}=5.12e17 \text{ kg/m}^3$ at approximately $9.85e11 \text{ s}$ in all spin-down cases. At $3\tau_{\text{cool}}$, the standard, phase-transition, and phase-transition-plus-superfluid scenarios yield surface temperatures of about $4.98e4 \text{ K}$, $5.32e4 \text{ K}$, and $2.36e5 \text{ K}$, respectively; phase_fraction reaches 1 only in the two transition scenarios. The mechanism therefore produces distinguishable model-internal thermal tracks, condi-

tional on the effective cooling, latent-heat, and compression proxies.). Supplementary mathematical-model PDF: quantitative_mathematical_models.pdf#Q2.

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